

Using a Spreadsheet To Fit Experimental pH Titration Data to a Theoretical Expression: Estimation of Analyte Concentration and K_a

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Titration continues to be a vital component of the undergraduate chemistry experience, and accordingly students are often introduced to several types including acid–base, redox, and complexation. A common acid–base pH titration experiment performed in quantitative analysis lab involves titrating V_a mL of an acid (ionization constant, K_a) of molar concentration F_a with a standardized solution of strong monoprotic base of molar concentration F_b . The equivalence point volume of the titration, V_{ep} , is estimated using one of several methods including (i) acid–base indicators, (ii) educated guess, (iii) first derivative or second derivative, or (iv) a Gran plot and F_a is subsequently determined using eq 1 (or an appropriately modified version for a polyprotic acid).

$$F_a = \frac{F_b V_{ep}}{V_a} \quad (1)$$

Moreover, a *point estimate* of pK_a can be obtained by realizing

$$pH = pK_a \text{ at } V_b = \frac{V_{ep}}{2} \quad (2)$$

This tried-and-true procedure can result in good estimations of F_a and K_a but it may be perceived as “old school”.

Although computer simulation of acid–base pH titrations has been discussed for over 30 years in this *Journal* (1–10) and fitting experimental titration data to theoretical expressions has been discussed in the research literature (11–22), it is interesting to note fitting experimental pH titration data has received relatively little pedagogical attention. In a 1993 article, focusing on the development of a single closed-form expression for the titration curve for a variety of titrations, de Levie (8a) indicates fitting experimental titration data to a derived theoretical expression is possible. Although a general procedure for this process is discussed in Appendix 2 of his article, it is in not the primary focus of the article. Ma and Tsang (23) discussed fitting experimental titration data to an arc tangent function, with attention focused on an accurate fit in the region of the equivalence point. While experimental titration data can be fit to such a function, no physical significance is associated with the four adjustable parameters of the arc tangent function. Eaker (24) has presented a calculator-based program that can be used to fit experimental weak monoprotic acid–strong base and strong monoprotic acid–strong base titration data to a piecewise equation. This single equation is a combination of three approximate equations each of which is appropriate in a specified region of the titration (i.e., before, at, and after the equivalence point). Eaker also discusses modeling titration data via a logistic function (a built-in regression option of some graphing calculators), although again no chemical significance is attached to the three determined parameters.

Most recently Kraft (25) has discussed the estimation of K_a values utilizing difference plots (a plot of an average number of dissociable protons, $\bar{n}_H$, bound to a weak acid H_nA versus pH). Kraft clearly indicates the utility of difference plots, yet the technique has several drawbacks, most notably the associated potential complexity. Difference plots are not typically discussed in undergraduate chemistry curriculum, and in fact the collected titration data (pH versus volume of added titrant) are not directly fit, but instead are used to calculate experimental $\bar{n}_H$ values, which are then fit to a theoretical expression. Although the technique described by Kraft results in the determination of K_a , the concentration of the titrated acid solution is not determined. Moreover, in some cases it may be necessary to add a known quantity of strong acid to the solution of weak acid prior to titration. This nonintuitive step ensures complete protonation of the weak acid, which corresponds to $\bar{n}_H = 1$.

Presented below is a technique that involves the direct fitting of weighted experimental acid–base pH titration data to a single theoretical expression derived via the “systematic treatment of equilibrium” (26). Fitting is accomplished using the Solver Add-in of the Microsoft Excel spreadsheet program. Solver is a powerful linear–nonlinear optimization software package produced by Frontline Systems Inc. (27) used to determine the value of parameters that “best” reproduce observed data. Benefits of this fitting procedure include:

- Derivation of a theoretical titration equation incorporates the systematic treatment of equilibrium and alpha fractions, two concepts often discussed in undergraduate analytical chemistry.
- As part of the fitting process students will be able to visualize how changing titration parameters affect the theoretical titration curve, that is, discern what happens to the shape of the theoretical titration curve when K_a is decreased, and understand the “agreement” between the theoretical and experimental curves.
- The presented methods and procedures are general and can be applied to any pH titration.
- Students are exposed to fitting and weighting data. These processes pervade essentially all fields of science, yet are often not stressed in undergraduate curricula. Readers interested in detailed descriptions of data fitting and weighting can consult appropriate references (28–32).
- K_a values and the analyte concentration resulting from the fitting process are determined quickly and with comparable accuracy to methods indicated above. Anfalt discusses precision and accuracy of several potentiometric endpoint detection methods and also arrives at this conclusion (11).

Procedure

To understand the process of fitting experimental titration data to a theoretical expression attention must be given to (i) the theoretical acid–base titration expressions (1–10), and (ii) a general method of fitting data using Excel (28–31a). Fortunately, both topics have been presented in this and other publications, and therefore interested readers are urged to consult the associated references.

Theoretical Titration Expression

Theoretical acid–base titration expressions can be expressed in several forms, any of which can be used in the fitting process. The form used in this work for the titration of V_a mL of a monoprotic acid HA (ionization constant K_a) of initial concentration F_a with a monoprotic base B (ionization constant K_b) of initial concentration F_b is given in eq 3¹ (a detailed derivation is included in the Supplemental Material^W),

$$V_b = \frac{(\alpha_{A^-})F_a V_a + \frac{K_w V_a}{[H_3O^+]} - [H_3O^+] V_a}{[H_3O^+] + (\alpha_{BH^+})F_b - \frac{K_w}{[H_3O^+]}} \quad (3)$$

where α is referred to as an alpha fraction (26)

$$\alpha_{A^-} = \frac{K_a}{[H_3O^+] + K_a} \quad (4)$$

$$\alpha_{BH^+} = \frac{K_b}{[OH^-] + K_b} = \frac{K_b [H_3O^+]}{K_w + K_b [H_3O^+]} \quad (5)$$

and K_w is the autoionization constant of water. Equation 3 is used to determine the volume of base V_b added to the receiving flask to produce a solution of a desired $[H_3O^+]$. In this form $[H_3O^+]$ is the independent variable and V_b the dependent variable (5). Although this is opposite of a pH titration and may seem bothersome, it creates no complication in the fitting process. Equation 3 can be applied to any titration of a monoprotic acid with any monoprotic base, and when a titration involves a strong acid or a strong base the corresponding equilibrium constant is set to an arbitrarily large value (~ 1000). Theoretical expressions for other acid–base titrations can be obtained following the procedure described in the Supplemental Material.^W

Fitting Data

Figures 1 and 2 are portions of an Excel spreadsheet used to fit experimental benzoic acid–sodium hydroxide pH titration data to equation 3. Columns D and E of Figure 1 are the experimental titration data and column A contains the calculated V_b corresponding to the pH of column E (columns

	A	B	C	D	E	F	G	H	I
1			$K_a = 6.11E-05$						
2		V_2 (ml) =	25.00						
3		F_2 (M) =	0.02470						
4		$K_b =$	1.00E+03						
5		F_1 (M) =	0.10815						
6		$K_w =$	1.00E-14						
7									
8		0.4657							
9	calc V_b	wt (cal V_b - exp V_b) ²	wt	exp V_b	pH	[H ⁺]	(α_{A^-})(F_2)	(α_{BH^+})(F_1)	$K_w/[H_3O^+]$
10	0.1466	0.0129	0.60	0.00	3.04	9.1201E-04	1.0815E-01	1.5517E-03	1.0955E-11
11	0.1844	0.0072	0.40	0.05	3.07	8.5114E-04	1.0815E-01	1.6552E-03	1.1749E-11
12	0.2099	0.0097	0.60	0.10	3.09	8.1283E-04	1.0815E-01	1.7277E-03	1.2303E-11
13	0.3152	0.0058	0.43	0.20	3.17	6.7608E-04	1.0815E-01	2.0482E-03	1.4791E-11
14	0.6025	0.0000	0.45	0.50	3.30	5.0119E-04	1.0815E-01	2.6863E-03	1.9963E-11
15	0.6494	0.0015	0.55	0.70	3.39	4.0730E-04	1.0815E-01	3.2299E-03	2.4547E-11
16	0.8494	0.0010	0.40	0.90	3.50	3.1623E-04	1.0815E-01	4.0014E-03	3.1623E-11
17	1.0135	0.0041	0.55	1.10	3.58	2.6303E-04	1.0815E-01	4.6581E-03	3.8019E-11
18	1.2660	0.0003	0.30	1.30	3.69	2.0417E-04	1.0815E-01	5.6914E-03	4.8978E-11
19	1.4172	0.0027	0.40	1.50	3.75	1.7783E-04	1.0815E-01	6.3188E-03	5.6234E-11
20	1.6333	0.0016	0.35	1.70	3.83	1.4791E-04	1.0815E-01	7.2231E-03	6.7608E-11
21	1.8350	0.0016	0.37	1.90	3.90	1.2589E-04	1.0815E-01	8.0735E-03	7.9433E-11

Figure 1. Screen capture of a portion of the Excel worksheet used to fit experimental benzoic acid–sodium hydroxide pH titration data to eq 3.

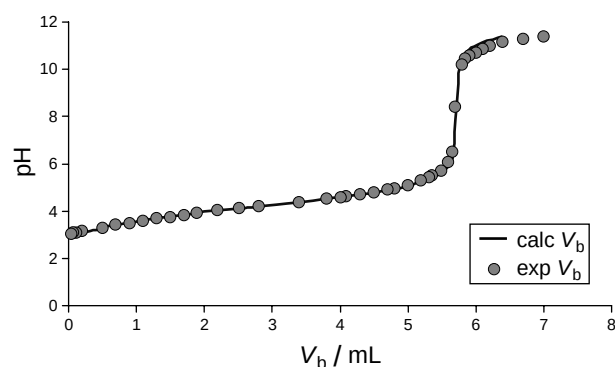


Figure 2. Comparison of the experimental benzoic acid–sodium hydroxide pH titration curve (points) and the best-fit theoretical pH titration curve (line). See Table 1 for the best fit values of K_a and benzoic acid concentration.

G–I are $[H_3O^+]$ -dependent terms needed to calculate V_b). Column B contains (cal V_b - exp V_b)² multiplied by the corresponding weight of column C (discussed below), and cell B8 is the sum of all values in column B. In addition to pH, V_b depends on the parameters given in cells B1–C6. V_a , F_b , and K_b are assumed to be known, and thus the goal of the fitting process is to determine values of K_a and F_a that best reproduce the experimental data (i.e., Solver is used to minimize the value of cell B8 by changing the values in cells C1 and C3).

Through empirical investigations the authors have found the following three-step process to be versatile method of fitting experimental titration data to a theoretical expression that has produced acceptable results for each of the titrations listed in Table 1. It must be stressed this process was developed through trial-and-error, and the authors acknowledge additional procedures can be used and may indeed be preferred. Students developing or exploring their own fitting procedures will have a greater appreciation of data fitting and weighting processes (the Supplemental Material^W includes additional discussion of this process).

Table 1. Results of Fitting Experimental Titration Data

Acid	Known Values		Data from Fitting	
	Molar Conc	pK _a	Molar Conc	pK _a
Hydrochloric	0.08699	—	0.08682	—
Benzoic	0.02381	4.19	0.02470	4.21
Acetic	0.1015	4.76	0.1004	4.76
Propionic	0.1038	4.82	0.1047	4.70
Maleic	0.05087	1.83	0.05066	1.60
		6.07		5.90
Oxalic	0.09997	1.23	0.1011	1.39
		4.19		3.83

NOTE: The experimental pK_a was obtained at 25°.

Weighting Data

In this pedagogical exercise weighing experimental data in the vicinity of equivalence points heavier than in other regions of the titration curve generally results in a determined F_a that is more accurate than when all experimental data are weighted equally. While other weighting schemes may be acceptable (21) a “derivative” weighting scheme has been utilized. As an example, the weight 0.80 in cell C12 of Figure 1 is calculated using cells D12–E13 as shown in eq 6.

$$wt = \frac{\Delta pH}{\Delta(\text{expt } V_b)} = \frac{3.09 - 3.17}{0.10 - 0.20} = 0.80 \quad (6)$$

Visual Fit

Approximate values of K_a and F_a are obtained by performing a “visual fit” of the experimental data. The plot of the fitting worksheet (i.e., Figure 2) is updated when the parameters in cells C1–C6 are changed, and thus the values of K_a and F_a are varied until the theoretical titration curve approximately coincides with the experimental data.

Solver

Solver is used to first determine the “best” values of K_a and F_a individually and then simultaneously. Typically each fit requires 1 to 2 iterations.

Results

Values of K_a and F_a resulting from the fit of pH titration of a 25.00 mL aliquot of several acids with standardized aqueous NaOH are shown in Table 1. The “known” molar concentration of each acid supplied in the table was determined by titrating a 25.00 mL aliquot with a standardized solution of NaOH to the phenolphthalein endpoint and is included only for purposes of comparison. The accuracy of the values resulting from the fitting process are typically better than those resulting from equivalence point determination methods mentioned above (for readers interested in exploring this aspect, experimental titration data have been included in the Supplemental Material^W). It should be noted the most accurate results were obtained when the pH electrode was filled with fresh 4 M KCl saturated AgCl solution prior to the titration, and the pH electrode or meter was cali-

brated with buffer solutions that encompassed a significant portion of the pH range covered during the titration. No attempt was made to maintain a constant ionic strength during the titration or to account for H₃O⁺ activity, although both are necessary to obtain the highest quality results.

Hazards

The hazards are identical to those associated with an acid–base titration. Proper precautions should be taken when preparing or using all acidic and basic solutions.

^WSupplemental Material

A derivation of a theoretical monoprotic acid–monoprotic base and diprotic acid–monoprotic base pH titration expressions, a description of least-squares fitting of data using the Excel Solver Add-in, Excel spreadsheets that can be used to determine K_a and F_a for monoprotic acid–monoprotic base and diprotic acid–monoprotic base pH titrations, and experimental acid–base pH titration data for the acids listed in Table 1 are available in this issue of *JCE Online*.

Note

- Equation 3 is equivalent to eq 12-10 (p 248) of ref 26.

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